

Session 13: Strings, Factors & Text — Pen-and-Paper Pair Exercise

PSY 410 | Data Science for Psychology

Name: _____ Date: _____

No laptop today? No problem. This handout lets you practice the same skills on paper. Work with a partner who has a laptop and compare your work at the end.

The data: messy_survey

You have messy survey responses:

id	major	comment
1	" PSYCHOLOGY"	"Great study!"
2	"biology"	"too long"
3	"Psychology"	" Very interesting "
4	"BIOLOGY"	"CONFUSING INSTRUCTIONS"
5	"sociology"	"I enjoyed this"

Notice the extra spaces and inconsistent capitalization.

The task (same as the slide exercise)

1. Clean `major` to lowercase with no extra spaces
2. Clean `comment` to title case with no extra spaces
3. Create a logical column `is_negative` that is TRUE if the comment contains "long" or "confusing" (case-insensitive)
4. Filter to only negative comments

Your pen-and-paper version

Step 1: Clean major. Apply `str_trim()` then `str_to_lower()`. Write the final value:

id	major (raw)	major (clean)
1	" PSYCHOLOGY"	
2	"biology"	
3	"Psychology"	
4	"BIOLOGY"	
5	"sociology"	

id	major (raw)	major (clean)
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Step 2: Clean comment. Apply `str_trim()` then `str_to_title()`. Write the final value:

id	comment (raw)	comment (clean)
1	"Great study!"	
2	"too long"	
3	" Very interesting "	
4	"CONFUSING INSTRUCTIONS"	
5	"I enjoyed this"	

Step 3: Detect negative comments. TRUE if the original comment contains "long" or "confusing" (case-insensitive):

id	comment	is_negative
1	"Great study!"	
2	"too long"	
3	" Very interesting "	
4	"CONFUSING INSTRUCTIONS"	
5	"I enjoyed this"	

Which rows survive `filter(is_negative)`? Rows _____ and _____

Step 4: Write the code. Fill in the blanks:

```
messy_survey |>
  mutate(
    major_clean = str_to_____ (str_____ (major)),
    comment_clean = str_to_____ (str_____ (comment)),
    is_negative = str_detect(
      comment,
      regex("_____|_____", ignore_case = _____)
    )
  ) |>
  filter(_____)
```

Check your work

Compare your cleaned values and code with your partner's screen. Do your answers match?